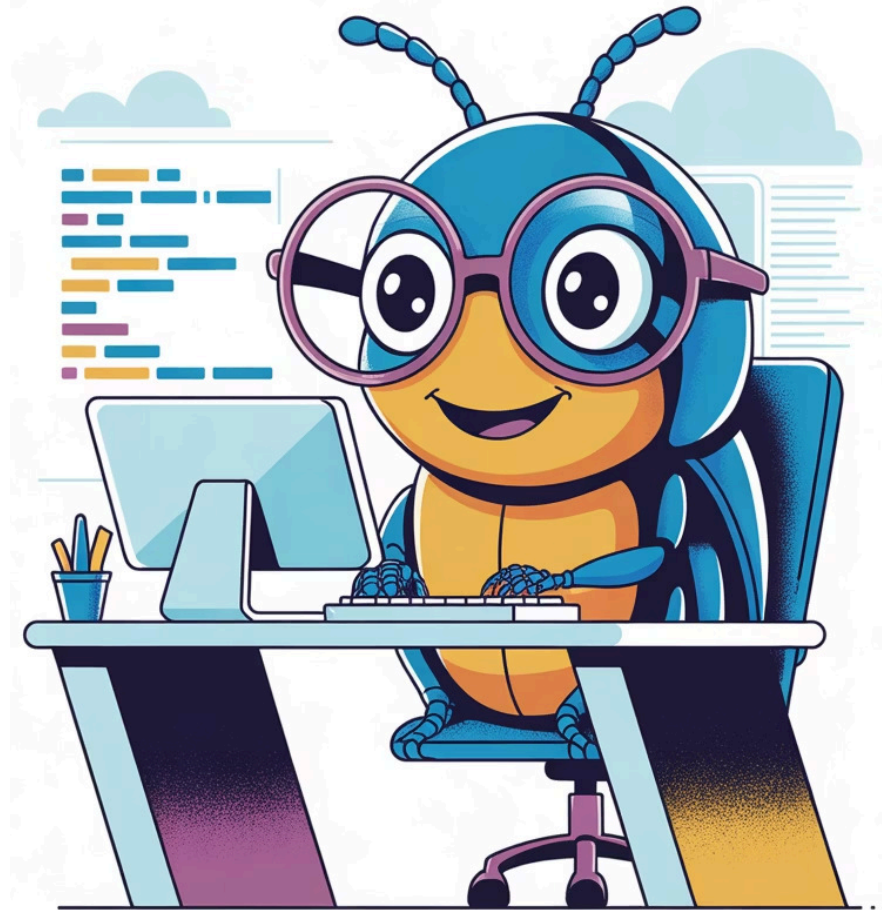
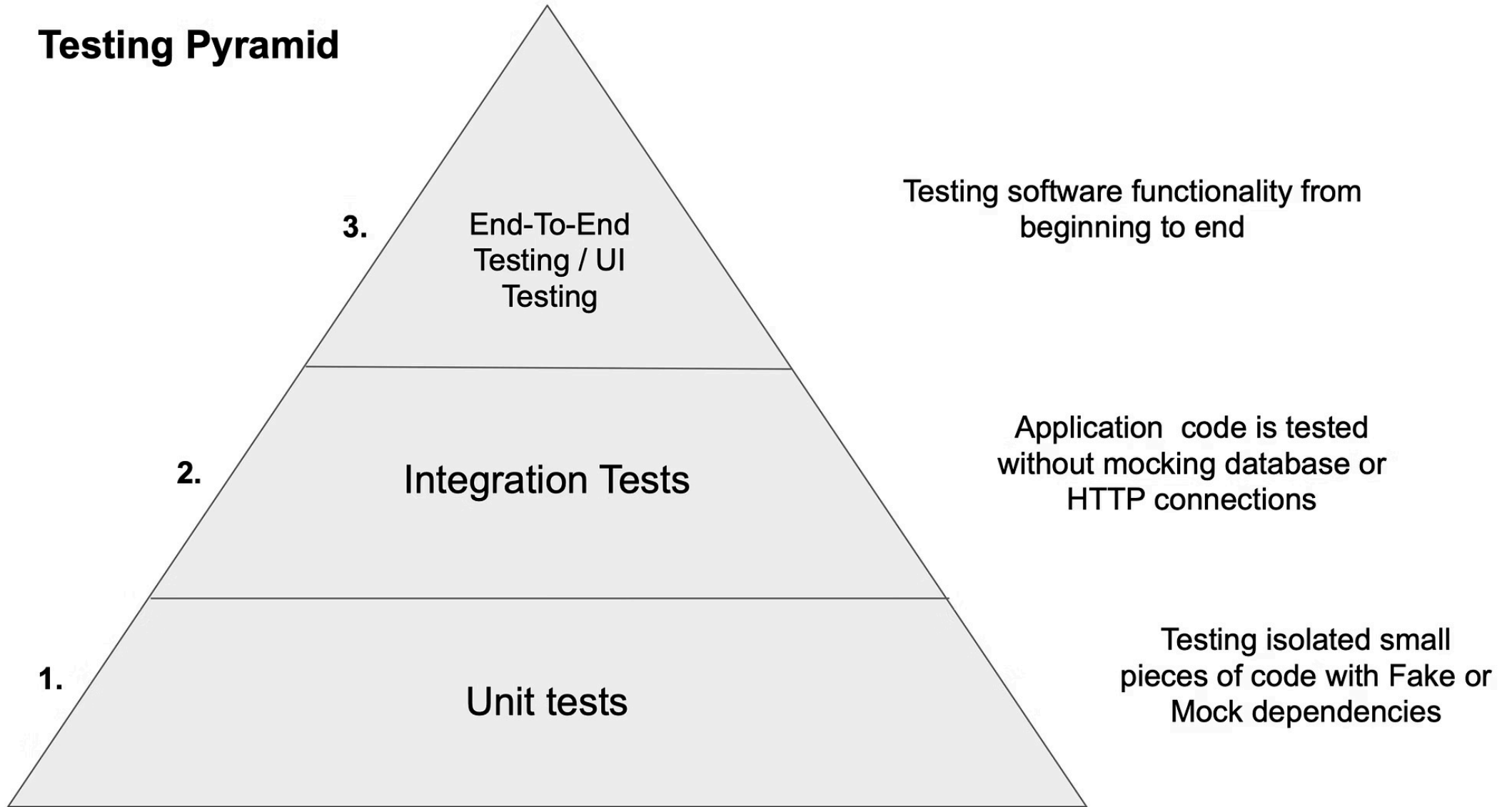


The Test Pyramid Overview

Applications typically require **three main categories** of **automated tests**



Testing Pyramid



The Test Pyramid Overview

Unit Tests Form the Base

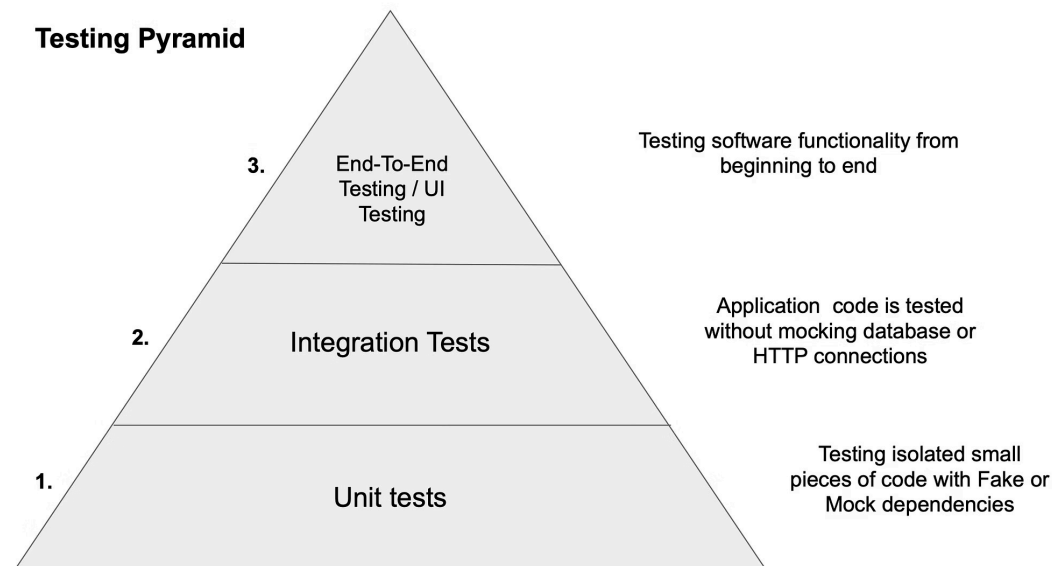
Unit tests form the base of the pyramid because they are most numerous

Unit tests **run early** and catch problems at the **smallest scope**

Speed Advantage

They are **fast** because they **avoid network calls** and **database access**

Their speed makes them ideal for **frequent execution during development**



Integration Tests

Real Dependencies

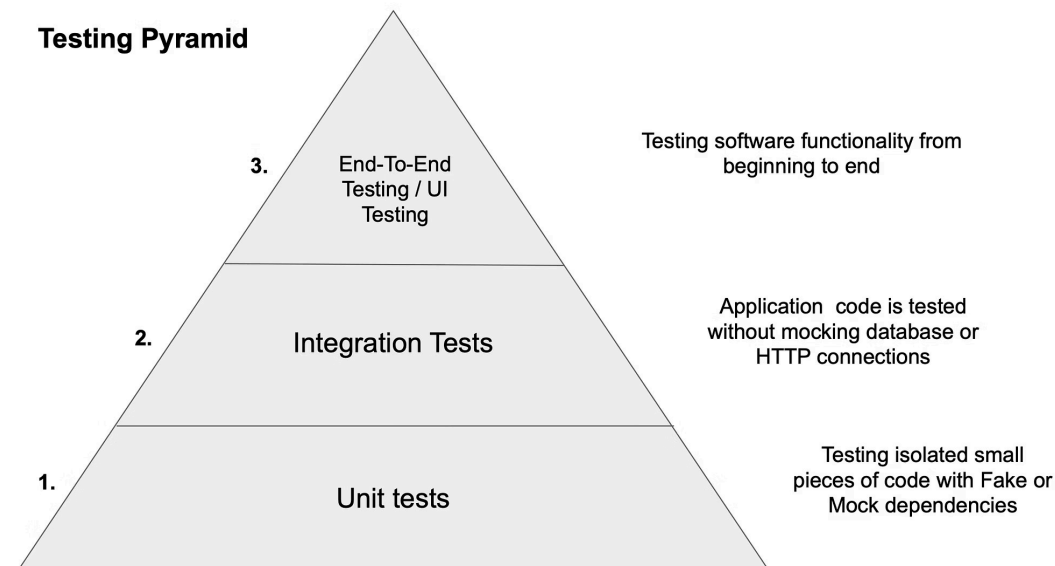
Integration tests **resemble unit tests** but use **real dependencies**

No mocks or **fakes** are used—**actual database** and **network** operations occur

These **tests validate correctness** when **components interact** with **external systems**

They **run slower** due to **I/O**, **remote calls**, and **real data operations**

Fewer **integration tests** are typically created **compared** to **unit tests**



End-to-End Testing

01

Full Workflow Validation

End-to-end (**E2E**) tests validate entire workflows from start to finish

02

Real User Simulation

They simulate **real user interactions**, such as **clicking UI buttons**

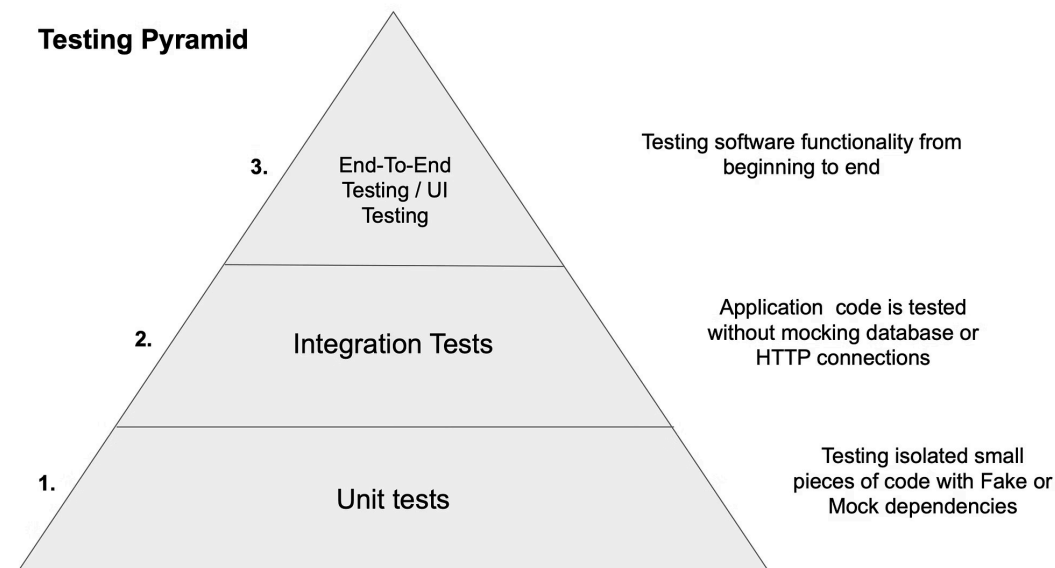
03

Comprehensive Testing

Useful for **testing mobile apps, web pages**, and **full system behavior**

- ❏ They are the **slowest and most resource-heavy** type of test

Only a small number of **E2E tests** are written due to **cost** and **complexity**



Conclusion

Fast Foundation

The test pyramid prioritizes fast, numerous unit tests at the base

Real-World Validation

Integration tests handle real-world interactions and compatibility

System Correctness

E2E tests ensure full-system correctness but should be used sparingly

Balanced Approach

A balanced test suite improves confidence while keeping feedback fast



Conclusion

Open Floor Discussion

- How do you currently balance unit, integration, and E2E tests in your projects?
- Which layer of the pyramid do you find most challenging to maintain?
- Have you experienced slowdowns caused by relying too heavily on E2E tests?

